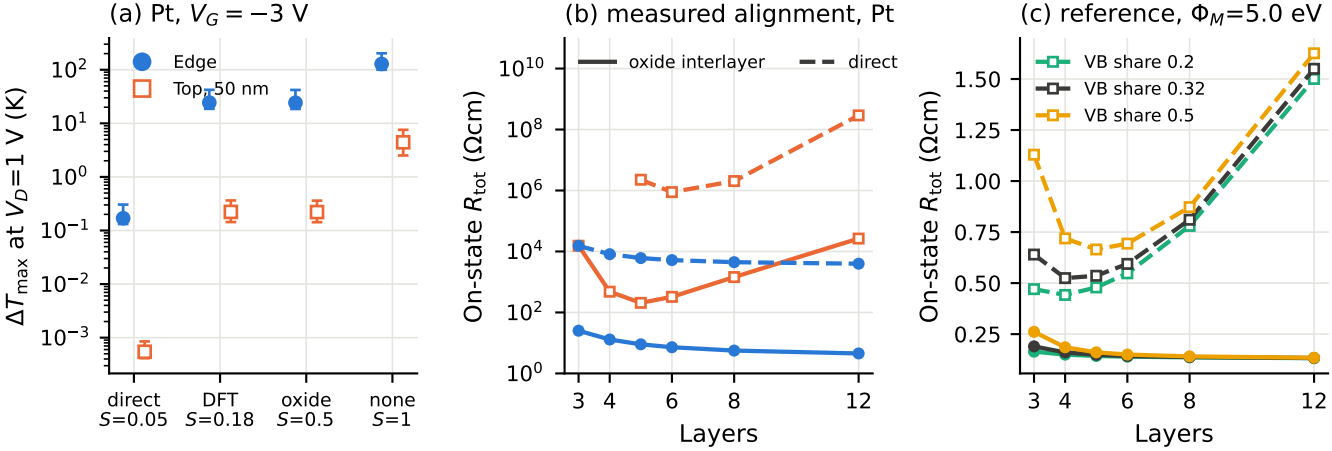
(a) Pt, $V_G = -3$ V
Edge
Top 50 nm
ΔT_{max} at V_D=1 V (K)
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
direct
S=0.05
DFT
S=0.18
oxide
S=0.5
none
S=1
(b) measured alignment, Pt
oxide interlayer
direct
On-state R_{tot} (Ωcm)
10^{10}
10^8
10^6
10^4
10^2
10^0
3
4
6
8
12
Layers
(c) reference, Φ_M=5.0 eV
VB share 0.2
VB share 0.32
VB share 0.5
On-state R_{tot} (Ωcm)
1.50
1.25
1.00
0.75
0.50
0.25
3
4
6
8
12
Layers